

Assignment - I

Write a computer program following the instructions given below:

1. ~~generate random~~
1. Initialize a coordinate (x_0, y_0) and put a dot at (x_0, y_0)
2. Generate a random no between 1 and 4, i.e. (1, 2, 3, 4) ~~5~~
3. ~~Generate~~ new coordinates following rule.
if random no is new coordinate updation rules.
1 $(0.8x + 0.1, 0.8y + 0.04)$
2 $(0.5x + 0.25, 0.5y + 0.4)$
3 $(0.355(x-y) + 0.266, 0.355(x+y) + 0.078)$
4 $(0.355(x+y) + 0.378, 0.355(y-x) + 0.434)$
4. Put a dot at new coordinate (x, y)
5. Repeat the process large no of time and generate the image.

Solution:

```
import random
```

```
from PIL import Image, ImageDraw
```

```
def generate_fractal_image(width, height, num_iterations):
```

```
    # 1. Initialize a coordinate (x0, y0) and put a dot at (x0, y0).
```

```
    x, y = 500, 500
```

```
    # Create a new blank black image
```

```
    img = Image.new('RGB', (width, height), 'black')
```

```
    draw = ImageDraw.Draw(img)
```

```
    # Put the initial dot
```

```
    draw.point((int(x), int(y)), 'white')
```

```
    # 5. Repeat the process a large number of times and generate the image.
```

```
    for _ in range(num_iterations):
```

```

# 2. Generate a random no between 1 and 4.
random_no = random.randint(1, 4)

if random_no == 1:
    x_new = 0.8 * x + 0.1 * width
    y_new = 0.8 * y + 0.04 * height
elif random_no == 2:
    x_new = 0.5 * x + 0.25 * width
    y_new = 0.5 * y + 0.4 * height
elif random_no == 3:
    x_new = 0.355 * (x - y) + 0.266 * width
    y_new = 0.355 * (x + y) + 0.098 * height
elif random_no == 4:
    x_new = 0.355 * (x + y) + 0.378 * width
    y_new = 0.355 * (y - x) + 0.434 * height

# Update the coordinates for the next iteration
x, y = x_new, y_new

# 4. Put a dot at new coordinate (x, y).
if 0 <= x < width and 0 <= y < height:
    draw.point((int(x), int(y)), 'white')

# Save the generated image
img.save('generated_image.png')
print("Image saved as generated_image.png")

# Set image parameters
image_width = 1000
image_height = 1000
iterations = 100000 # A large number of iterations to get a good result

# Run the function to generate the image
generate_fractal_image(image_width, image_height, iterations)

```

Image:

